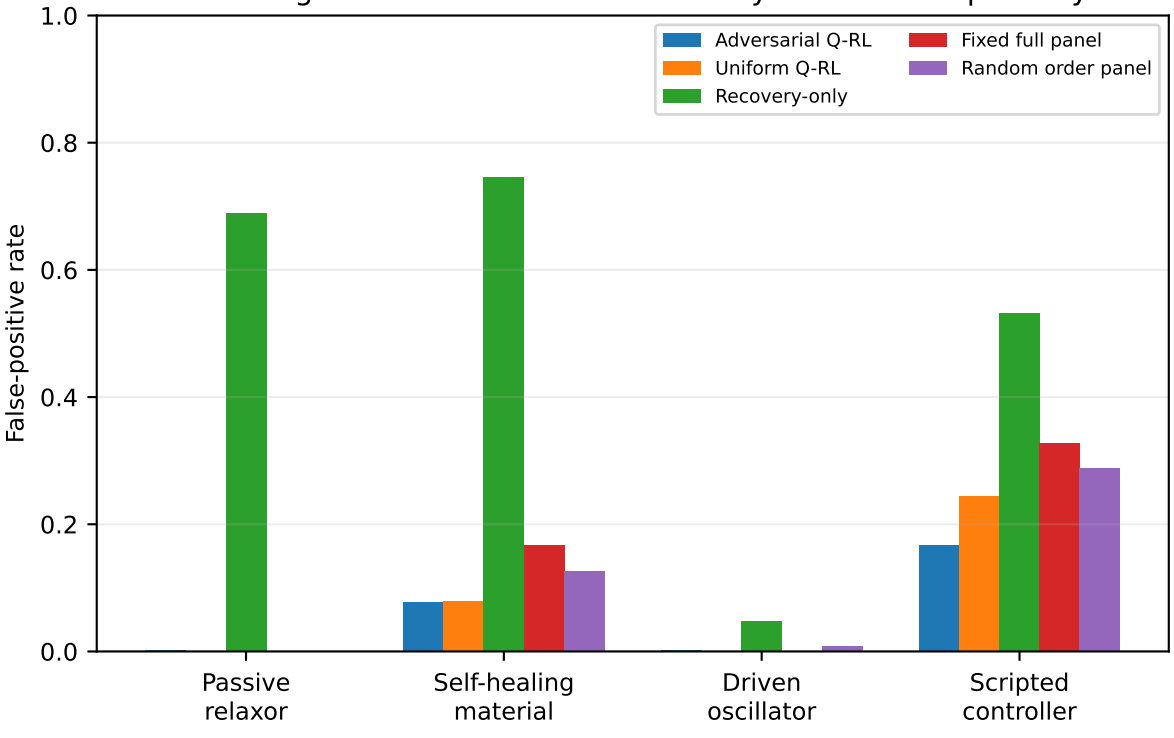
Hard-negative audit: each abiotic family evaluated separately
False-positive rate
1.0
0.8
0.6
0.4
0.2
0.0
Adversarial Q-RL
Uniform Q-RL
Recovery-only
Fixed full panel
Random order panel
Passive relaxor
Self-healing material
Driven oscillator
Scripted controller